

# llmXive Automated Review of Playful Agentic Robot Learning

*llmXive automated review panel*

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**Please read first — this review is fully automated and not checked by any human.** It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

## How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

**This paper's panel (9 lenses):** Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

## Claim Accuracy

*Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong quantitative claims regarding performance improvements across

multiple benchmarks (LIBERO-PRO, MolmoSpaces, RoboSuite, and real-world). The primary factual accuracy concern lies in the interpretation of the “Cross-Environment Transfer” results in Section 5.2 and Table 3.

The abstract and Section 1 state that skills transfer to RoboSuite, “improving performance by 8.9 points.” Table 3 confirms the *average* improvement is indeed +8.9 pp (40.3% to 49.1%). However, the table explicitly shows that the “Two-arm handover” task *decreased* in performance by 4.0 pp (from 24.0% to 20.0%). While the average claim is mathematically correct, the phrasing “improving performance” without qualification could mislead a reader into believing the transfer was universally positive. The text should be slightly refined to specify that the *average* performance improved, or explicitly note that while most tasks saw gains, some (like handover) did not.

Additionally, in Section 3.2, the definition of the Competence Frontier  $\mathcal{F}(\tau)$  relies on  $\bar{r}(\tau)$ , described as the “average Wilson-bounded empirical success rate.” The Wilson score interval is a method for estimating a confidence interval for a binomial proportion, not a standard point estimator for the mean itself. If  $\bar{r}$  is the lower bound of the Wilson interval, the formula  $4\bar{r}(1 - \bar{r})$  behaves differently than if  $\bar{r}$  is the raw empirical mean. The claim that this function “peaks at  $\bar{r} \approx 0.5$ ” is only strictly true if  $\bar{r}$  is the raw mean. If  $\bar{r}$  is a lower bound (which is typically  $< 0.5$  for low counts), the peak of the function shifts. The authors must clarify whether  $\bar{r}$  is the raw mean or the Wilson lower bound to ensure the mathematical claim about the “Goldilocks” peak is accurate.

Finally, the citation `fu2026capx` refers to a paper dated 2026. While preprints often have future-dated arXiv IDs or submission years in metadata, citing a 2026 paper as a definitive source for “CaP-X shows multi-turn feedback benefits” in a 2024/2025 context requires verification that the work is indeed publicly available and established under that specific citation key. If the paper is not yet published or available, the citation year or status should be adjusted to reflect the actual availability.

### Action items.

- **[writing]** The abstract and Section 1 claim RoboSuite transfer improves performance by 8.9 points. Table 3 shows the average gain is +8.9 pp (40.3% to 49.1%), but the ‘Two-arm handover’ task shows a -4.0 pp decrease. The text should clarify that the 8.9 pp is an average across tasks, not a uniform improvement, to avoid implying every task improved.
- **[science]** Section 3.2 defines the Competence Frontier  $\mathcal{F}(\tau)$  as peaking at  $\bar{r}$  approx 0.5. However, the text states  $\bar{r}$  is the ‘average Wilson-bounded empirical success rate’. Wilson bounds are typically used for confidence intervals on proportions, not as a direct point estimate for the mean success rate in a formula. The authors should clarify if  $\bar{r}$  is the raw empirical mean or the lower bound of the Wilson interval, as this changes the mathematical interpretation of the ‘peak’ at 0.5.
- **[writing]** The bibliography lists ‘fu2026capx’ (CaP-X) with a 2026 publication year. As this is a preprint review, citing a future-dated paper (2026) as an established reference for ‘CaP-X shows multi-turn feedback benefits’ is factually premature unless the paper is already publicly available under that specific citation key. The authors should verify the citation

status or adjust the year to the actual arXiv submission date.

- **[writing]** Table 3 reports ‘Two-arm lifting’ success as 5/50 (10.0%) for CaP-Agent0 and 17/50 (34.0%) for RATs. The delta is listed as +24.0 pp. However,  $34.0 - 10.0 = 24.0$ . The text in Section 5.2 claims ‘Notable gains: ... two-arm lifting (+24.0 pp)’. This is mathematically correct, but the table also lists ‘Two-arm handover’ with a -4.0 pp delta. The text should ensure the ‘notable gains’ list does not inadvertently imply all transfer tasks improved, given the handover failure.

## Figure Critic

*Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_figure_critic.md`

### Figure 1

The figure effectively displays qualitative comparisons of robotic tasks between two methods, but the caption is insufficient as it fails to describe the specific tasks or the methods being compared.

### Figure 2

The figure displays qualitative real-world robot tasks but fails to visually support the caption’s claim of ‘sim-to-real transfer’ by omitting the necessary simulation comparisons or baselines.

### Figure 3

Figure 3 is a clear and effective teaser figure that visually summarizes the paper’s core contribution. The layout logically separates the ‘Play-Time’ learning phase from the ‘Test-Time’ application phase, and the overlaid code snippets effectively illustrate the ‘Code-as-Policy’ mechanism described in the caption.

### Figure 4

Figure 4 is a clear and well-structured pipeline diagram that effectively visualizes the RATs architecture described in the caption. All stages (Task Proposing, Planning, Execution, Verification) are clearly labeled with representative examples, and the feedback loops for success and failure are explicitly annotated.

## Figure 5

Figure 5 is a clear, self-contained diagram illustrating the task proposal process. All components, including the scene input, scoring logic, candidate ranking, and selected outcome, are visually distinct and well-labeled, effectively supporting the caption’s description of an example trace.

## Figure 6

Figure 6 is a clear and well-constructed bar chart that effectively visualizes the distribution of play objectives. The axes are labeled with units, the data values are explicitly annotated on the bars, and the caption accurately describes the data source and context.

## Figure 7

The figure effectively visualizes the growth of the skill library and failure memory, but the stacked area chart format in the left panel obscures the specific values for individual skill categories (Verified, Experimental, Deprecated), making precise comparison difficult. Additionally, the axis label ‘Learned helpers’ does not perfectly align with the caption’s terminology.

## Figure 8

The heatmap effectively visualizes the distribution of skill calls, but the caption’s claim about column aggregation contradicts the raw counts shown, and the dual-number cell format lacks explicit visual distinction between counts and percentages.

## Figure 9

Figure 9 effectively visualizes the play-to-evaluation transfer lineage described in its caption. The progression from a failed iteration to a successful one, and finally to the evaluation task, is clearly illustrated with corresponding code snippets and visual states.

## Figure 10

Figure 10 effectively contrasts the ‘CaP-Agent0 Direct Synthesis’ approach with ‘RATS with learned skills’ using side-by-side code excerpts and visual outcomes. The figure clearly demonstrates the failure of the direct synthesis method versus the success of the RATS method, supported by a detailed legend and clear visual cues (red X vs. green checkmark) that align with the caption’s claim about reducing brittle low-level reasoning.

## Figure 11

Figure 11 effectively communicates the qualitative comparison between the CaP-Agent0 and RATS systems across three distinct robotic tasks. The layout is clear, with visual examples paired with code snippets that highlight the specific architectural differences (inline computation vs. skill reuse) described in the caption.

Figure 12

The figure fails to visually support the caption’s specific claims about LIBERO-to-RoboSuite transfer and skill reuse, showing only a generic success state without the necessary comparative or source-contextual elements.

#### Action items.

- **[writing]** Figure 1: The caption ‘Qualitative comparisons in simulation’ is too generic and does not describe the specific tasks (e.g., ‘open the drawer’, ‘turn on the stove’) or the methods being compared (CaP-Agent0 vs. RATs) shown in the image.
- **[writing]** Figure 1: The task labels on the left include dataset names in parentheses (e.g., ‘(MolmoSpaces)’, ‘(LIBERO-PRO)’), but the caption does not define these terms or explain their relevance to the simulation environment.
- **[science]** Figure 2: The caption claims to show ‘sim-to-real transfer’ results, but the image displays four rows of real-world robot execution sequences (Place Cube, Swap Cubes, Close/Open Drawer) without any corresponding simulation baselines or side-by-side comparisons to demonstrate the transfer.
- **[writing]** Figure 2: The image contains no figure number, title, or caption text embedded within the visual itself, relying entirely on external placement which is not visible in the provided render.
- **[science]** Figure 7: The left panel (‘Skill library’) is a stacked area chart where the y-axis represents the cumulative total of all categories. However, the legend labels (‘Verified’, ‘Experimental’, ‘Deprecated’) imply these are separate series. The visual representation makes it impossible to read the specific count of ‘Verified’ skills (the bottom dark band) without estimating the height of the band itself, which is obscured by the bands above it. A line plot or bar chart would be clearer for comparison.
- **[writing]** Figure 7: The caption states the figure reports ‘verified/experimental/deprecated skill counts’, but the left plot’s y-axis is labeled ‘Learned helpers’. While likely synonymous in context, the terminology mismatch between the axis label and the caption/legend creates ambiguity.
- **[science]** Figure 8: The caption states ‘Each column aggregates 100 evaluation trials,’ but the ‘Pick’ column sums to 846 calls (714+0+97+139+0), which is inconsistent with the stated trial count or implies a misunderstanding of how ‘calls’ vs ‘trials’ are aggregated.
- **[writing]** Figure 8: The heatmap cells contain two numbers (count and percentage) stacked vertically without explicit labels or delimiters, which forces the reader to infer which is which based on the caption; adding a legend or distinct formatting (e.g., bold for count) would improve clarity.
- **[science]** Figure 12: The caption claims the figure shows ‘LIBERO-to-RoboSuite skill transfer’ and that RATS succeeds by ‘reusing skills selected from a LIBERO-derived library,’ but the figure itself contains no visual evidence of the LIBERO source, the library contents, or the

specific skills being reused; it only shows a generic ‘Success’ state.

- **[writing]** Figure 12: The figure is a single static image showing a ‘Success’ state, but the caption implies a comparison (‘direct code synthesis fails while RATS succeeds’) that is not visually represented in the provided image (e.g., no ‘Failed’ panel or side-by-side comparison).

## Jargon Police

*Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_jargon\_police.md

The manuscript relies heavily on domain-specific jargon and undefined acronyms, which creates a barrier for readers outside the immediate sub-fields of robotics and intrinsic motivation.

First, the term **Code-as-Policy** is introduced in the Abstract and Introduction as a central concept but is never explicitly defined. While the title suggests it involves code, the specific mechanism (using LLMs to synthesize executable programs for control) should be briefly explained upon first mention to aid non-specialists.

Second, the acronyms **VLM** (Vision-Language Model) and **VLA** (Vision-Language Action) are used frequently (e.g., Abstract, Section 1, Limitations) without being spelled out at their first occurrence. Standard academic practice requires defining these immediately.

Third, the abbreviation **pp** for “percentage points” appears in the Abstract (“+20.6 pp”) and throughout Section 4. This is informal shorthand that should be written out in full for clarity in a formal publication.

Fourth, in Appendix A.1, the term **BDDL** is used to describe the environment generation format. This is a specific planning language acronym that should be defined (e.g., “Basic Domain Definition Language (BDDL)”) for readers unfamiliar with automated planning.

Finally, the phrase “**Wilson-bounded** empirical success rate” in Section 3.2 is dense jargon. While the Wilson interval is a standard statistical tool, the phrasing obscures the meaning. A clearer phrasing would be “success rate bounded by the Wilson interval” or “Wilson lower bound of the success rate.”

Addressing these points will significantly improve the accessibility of the paper without altering its scientific content.

### Action items.

- **[writing]** The manuscript relies heavily on domain-specific jargon and undefined acronyms, which creates a barrier for readers outside the immediate sub-fields of robotics and intrinsic motivation. First, the term Code-as-Policy is introduced in the Abstract and Introduction as a central concept but is never explicitly defined. While the title suggests it involves code, the specific mechanism (using LLMs to synthesize executable programs for control) should be briefly explained upon first mention to aid no

## Logical Consistency

*Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_logical\_consistency.md

The paper presents a compelling framework for ‘Playful Agentic Robot Learning’ using the RATs system. The logical flow from the problem statement (task-driven limitations) to the proposed solution (self-directed play with skill distillation) is generally sound. However, there are specific areas where the causal claims and definitions require tighter logical consistency to fully support the conclusions.

First, the claim of transfer “without finetuning” (Abstract, Sec 1) creates a logical tension with the “Plug-and-Play” mechanism described in Sec 3.5 and Appendix E002. The system injects a frozen library of code skills into the context of CaP-Agent0. If CaP-Agent0 was not originally trained to utilize such a dynamic, external skill library, the performance gain implies that the agent is effectively performing a form of in-context learning or zero-shot adaptation that functions similarly to fine-tuning. The term “without finetuning” is technically accurate regarding gradient updates but may be misleading regarding the *computational* and *contextual* adaptation required. The authors should clarify whether the base agent’s architecture inherently supports this dynamic library injection or if the “plug-and-play” process involves a non-trivial adaptation step that should be acknowledged as a form of zero-shot fine-tuning.

Second, the causal link between the “Goldilocks” task selection rule and performance gains needs stronger evidence. The formula in Sec 3.2 explicitly targets tasks with a success rate near 0.5. While the ablation in Table 4 shows “Curious Play” (32.3%) outperforms “Random Play” (24.7%), the paper does not explicitly verify that the tasks *actually selected* by the Goldilocks rule were indeed those with success rates near 0.5, nor does it prove that the *specific* selection of these “just-right” tasks was the primary driver of the gain, rather than simply the increased diversity or volume of practice. A breakdown of the empirical success rates of the *proposed* vs. *selected* tasks would strengthen the logical argument that the Goldilocks heuristic is the specific mechanism of success.

Finally, the compute-matched comparison in Appendix E002 (Table 6) attempts to isolate the value of play-time compute versus test-time retries. The logic holds that play-time compute is more effective, but the comparison relies on the assumption that the 30M tokens spent in play are strictly “extra” cost. If the play phase is viewed as a training phase, a more rigorous logical control would be to compare the RATs system against a baseline that also undergoes a 50-iteration “training” phase (even if random or naive) to ensure the gain is due to the *curiosity* mechanism and not just the *total volume of compute* applied to the problem. The current framing risks conflating the efficiency of the *strategy* with the sheer amount of *compute* applied.

### Action items.

- **[writing]** The paper presents a compelling framework for ‘Playful Agentic Robot Learning’ using the RATs system. The logical flow from the problem statement (task-driven limitations)



to the proposed solution (self-directed play with skill distillation) is generally sound. However, there are specific areas where the causal claims and definitions require tighter logical consistency to fully support the conclusions. First, the claim of transfer “without finetuning” (Abstract, Sec 1) creates a logical tension

## Overreach

*Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_overreach.md

The paper makes several claims regarding the autonomy and generalization of the \textsc{RATs} system that appear to exceed the evidence provided in the text.

First, the Abstract and Section 5.3 claim that skills transfer to real-world tasks “without finetuning.” While the authors state they do not finetune the LLM, the claim implies a level of zero-shot generalization that is not fully supported. The real-world evaluation (Table 3) shows modest gains (+8.8 pp), but the paper does not clarify if the low-level primitives (e.g., grasp controllers, inverse kinematics) used by the generated code were also frozen or if they required environment-specific tuning. If the primitives were tuned for the real robot, the claim of “without finetuning” is misleading regarding the *system’s* ability to transfer, as the transfer relies on pre-tuned components. The distinction between “no LLM finetuning” and “no system finetuning” must be precise.

Second, the characterization of the learning process as “self-directed play” (Abstract, Introduction) overstates the autonomy of the Task Proposer. The Appendix (Sec. Appendix Prompt Templates) reveals that the Task Proposer is explicitly conditioned on “KNOWN ENV LIMITATIONS” and “PICK-PRIMITIVE RELIABILITY” lists. This indicates that the “play” is not entirely self-directed or open-ended but is heavily constrained by human-defined priors and safety constraints. The paper should temper the “self-directed” claim to reflect that the agent operates within a bounded, human-curated search space, rather than discovering entirely novel interaction modes from scratch.

Finally, the “Limitations” section vaguely states that “Improper skill reuse can hurt performance.” However, Table 3 (Cross-environment transfer) provides a concrete counter-example: the “Two-arm handover” task sees a performance *drop* from 24.0% to 20.0% when using \textsc{RATs} skills. This is not just a theoretical risk but an observed failure mode of the proposed method. The paper should explicitly analyze this negative transfer case in the main text or discussion rather than burying it in a generic limitation, as it challenges the robustness of the “play” curriculum in generating universally beneficial skills.

### Action items.

- **[writing]** The paper makes several claims regarding the autonomy and generalization of the \textsc{RATs} system that appear to exceed the evidence provided in the text. First, the Abstract and Section 5.3 claim that skills transfer to real-world tasks “without finetuning.”



While the authors state they do not finetune the LLM, the claim implies a level of zero-shot generalization that is not fully supported. The real-world evaluation (Table 3) shows modest gains (+8.8 pp), but the paper does not clarify if

## Safety Ethics

*Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_safety\_ethics.md

The manuscript addresses safety and ethics primarily through a brief, high-level statement in Section 1 (“Safety Considerations”) and a mention of a “Quality Checker” agent in the methodology. While the authors cite ISO 10218 and ISO 15066, the review finds the current treatment of safety insufficient for a system involving autonomous, self-directed code generation and physical robot interaction.

First, the “Safety Considerations” section (Section 1) is too generic. It lists compliance standards but fails to explain the specific mechanisms implemented in the \textsc{RATs} framework to ensure these standards are met during the “play” phase. Since the agent autonomously proposes tasks and writes code, there is a non-trivial risk of generating policies that violate physical safety constraints (e.g., excessive force, unsafe trajectories). The review requires a detailed explanation of how the “Quality Checker” agent (Section 4.2) specifically validates code against safety constraints before execution, and what hard-coded safety limits (e.g., joint velocity limits, force thresholds) are enforced by the simulation or real-world controller.

Second, the transition to real-world evaluation (Section 5.4) raises significant safety concerns. The paper reports a 38.8% success rate on a real robot using skills learned in simulation. However, it does not clarify if the “play” phase itself was ever conducted on the physical robot. If “play” was restricted to simulation, the authors must explicitly state this and discuss the “sim-to-real gap” regarding safety. If “play” was conducted on the real robot, a rigorous risk assessment is missing. Autonomous exploration on physical hardware carries inherent risks of damage to the robot, the environment, or injury to nearby personnel. The manuscript must detail the specific safeguards (e.g., human-in-the-loop supervision, virtual fencing, emergency stop protocols) used during any real-world experimentation.

Finally, the “Limitations” section acknowledges that “improper skill reuse can hurt performance” but does not address the safety implications of such failures. A learned skill that is statistically successful in simulation but fails in a specific real-world context could lead to hazardous behavior (e.g., dropping a heavy object, colliding with a human). The authors should expand this section to discuss the potential for safety-critical failures arising from the autonomous learning process and outline any proposed mitigation strategies, such as runtime monitoring or human verification of new skills before deployment.

The paper is otherwise scientifically sound, but these safety and ethical gaps must be addressed to ensure the responsible deployment of such autonomous systems.

**Action items.**

- **[writing]** The manuscript addresses safety and ethics primarily through a brief, high-level statement in Section 1 (“Safety Considerations”) and a mention of a “Quality Checker” agent in the methodology. While the authors cite ISO 10218 and ISO 15066, the review finds the current treatment of safety insufficient for a system involving autonomous, self-directed code generation and physical robot interaction. First, the “Safety Considerations” section (Section 1) is too generic. It lists compliance standards

**Scientific Evidence**

*Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_scientific\_evidence.md

The paper presents a compelling framework for “Playful Agentic Robot Learning” with substantial reported gains across LIBERO-PRO (+20.6 pp), MolmoSpaces (+17.0 pp), and transfer tasks. However, the scientific evidence supporting these claims lacks necessary statistical rigor.

The primary concern is the absence of variance metrics. Tables 1, 2, and 3 report single-point success rates (e.g., 43.8% vs 23.2%) without standard deviations or confidence intervals. In agentic systems involving LLMs and stochastic environments, performance can fluctuate significantly based on random seeds. Without reporting results averaged over multiple seeds (e.g., 3-5) with standard deviations, it is impossible to determine if the observed improvements are statistically significant or artifacts of a specific lucky/unlucky seed configuration. The “Reproducibility Checklist” mentions fixed seeds, but this does not substitute for variance analysis in the main results.

Furthermore, the “compute-matched” comparison in Table 5 (Section 5.3) is methodologically weak. The baseline (CaP-Agent0 with 15 turns) is compared against the proposed method, but the statistical robustness of the 15-turn baseline is not established. If the baseline was evaluated on a single seed, the claim that “proactive play-time computation is more effective” is not fully supported by the evidence provided.

Finally, the ablation studies (Table 4) show clear trends, but again lack error bars. The difference between “Random Play” (24.7%) and “Curious Play” (32.3%) is notable, but without variance, the significance of the “Curiosity” component’s contribution remains an assumption rather than a proven fact. The authors should re-run experiments with multiple seeds and report mean  $\pm$  std dev to substantiate the robustness of their central claims.

**Action items.**

- **[science]** Report statistical significance (e.g., p-values, confidence intervals, or standard deviations) for the reported percentage point improvements (e.g., +20.6 pp in LIBERO-PRO). The current tables present single-point averages without variance metrics, making it impossible to assess if gains are robust or due to random seed variance.

- **[science]** Clarify the random seed protocol. The reproducibility checklist mentions ‘Fixed seeds (42)’, but LLM-based agents often exhibit high variance even with fixed seeds due to non-deterministic sampling or environment stochasticity. Report results averaged over multiple seeds (e.g., 3-5) with standard deviation to validate the stability of the +17.0 pp gain in MolmoSpaces.
- **[science]** Provide a more rigorous control for the ‘compute-matched’ baseline in Table 5. The comparison between a 15-turn baseline and the play-learned model needs to explicitly account for the variance in the 15-turn baseline. If the 15-turn baseline was only run once, the comparison is statistically weak.

## Statistical Analysis

*Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_statistical\_analysis.md

The manuscript presents compelling empirical results for the \textsc{RATs} framework, but the statistical rigor of the analysis requires strengthening to support the quantitative claims.

First, the primary results tables (Table 1: LIBERO-PRO, Table 2: MolmoSpaces, Table 3: RoboSuite/Real-World) report success rates as single-point percentages (e.g., 43.8%, 38.0%). In experimental robotics and machine learning, reporting point estimates without measures of uncertainty (standard deviation, standard error, or 95% confidence intervals) is insufficient. Given the stochastic nature of LLM-based agents and simulation environments, the variance across trials is critical. For instance, in Table 1, the improvement from 23.2% to 43.8% is substantial, but without error bars or confidence intervals, it is impossible to determine if this difference is statistically significant or if the observed gains could be attributed to random fluctuation in the specific random seeds used.

Second, the manuscript claims significant improvements across multiple sub-categories (Object, Goal, Spatial in LIBERO-PRO; Open, Close, Pick, PnP in MolmoSpaces) without addressing the issue of multiple comparisons. When testing multiple hypotheses (e.g., 6 sub-splits in LIBERO-PRO), the family-wise error rate increases. The authors should explicitly state whether they applied corrections (such as Bonferroni or Holm-Bonferroni) to the p-values or if they treated each sub-task as an independent experiment. If no correction was applied, the discussion should acknowledge the increased risk of Type I errors.

Third, the statistical methodology for the “compute-matched” comparison in Appendix Table \ref{tab:compute\_matched\_comparison} needs clarification. The comparison between a 10-turn baseline and a 15-turn baseline involves different computational budgets. While the authors attempt to match token costs, the statistical validity of comparing these distributions requires a clear statement on the test used (e.g., Mann-Whitney U test for non-normal distributions, or paired t-test if assumptions are met). The text mentions “modest gains” but does not provide a p-value or effect size (e.g., Cohen’s d) to quantify the magnitude of the difference between the

23.2% and 26.0% results.

Finally, the definition of the experimental unit in Section \ref{subsec:exp-details} and Appendix \ref{sec:appendix\_evaluation\_benchmarks} states “ten initial-state trials per task.” It is crucial to clarify if these 10 trials are independent runs with different random seeds or 10 runs with a fixed seed but different initial states. If the latter, the degrees of freedom for any statistical test must reflect the number of unique initial states, not the total number of rollouts, to avoid pseudoreplication. The current text implies 600 total rollouts for LIBERO-PRO, but the statistical power depends on the independence of these samples.

To improve the paper, the authors should re-run the analysis to include 95% confidence intervals (via bootstrapping or standard error propagation) for all mean success rates in the main tables. Additionally, a brief “Statistical Analysis” subsection should be added to the Experimental Results section detailing the specific tests used, the handling of multiple comparisons, and the justification for the chosen significance threshold.

#### Action items.

- **[writing]** The manuscript presents compelling empirical results for the \textsc{RATs} framework, but the statistical rigor of the analysis requires strengthening to support the quantitative claims. First, the primary results tables (Table 1: LIBERO-PRO, Table 2: MolmoSpaces, Table 3: RoboSuite/Real-World) report success rates as single-point percentages (e.g., 43.8%, 38.0%). In experimental robotics and machine learning, reporting point estimates without measures of uncertainty (standard deviation, standar

## Writing Quality

*Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_writing\_quality.md

The manuscript demonstrates a high level of technical sophistication, and the writing is generally clear and concise. The structure is logical, and the use of bolding for key terms and concepts aids in quick comprehension. However, there are a few areas where the flow and grammatical consistency could be improved to enhance readability.

In Section 2 (Related Work), the transition between the bolded header “Play and Curiosity” and the first sentence of the paragraph is slightly abrupt. The sentence “Play provides a self-directed curriculum...” is a strong statement, but it could be better integrated with the header by adding a brief introductory clause or rephrasing to create a smoother flow.

In Section 3.2 (Task Proposer Team), the explanation of the Competence Frontier formula is dense. The sentence “where  $\bar{r}$  is the average Wilson-bounded empirical success rate of required skills. Peaks at  $\bar{r}$  approx 0.5.” is a bit fragmented. Consider combining these into a single, more fluid sentence or using a colon to introduce the explanation, which would improve the readability of the mathematical description.

In Section 5.2 (Cross-Environment Transfer), the phrase “Notable gains: cube lifting (+16.0

pp), two-arm lifting (+24.0 pp).” is a sentence fragment. While this style is common in technical writing for brevity, it stands out against the otherwise complete sentences in the paragraph. Integrating this into a full sentence (e.g., “Notable gains include a 16.0 pp improvement in cube lifting and a 24.0 pp improvement in two-arm lifting.”) would maintain grammatical consistency.

Finally, in the Appendix, Section A.4 (Agent Prompts), the transition between the ‘SYSTEM PROMPT’ and ‘USER TEMPLATE’ sections could be clarified. Adding a brief explanatory sentence before the code block would help guide the reader through the structure of the prompt templates, ensuring that the purpose of each section is immediately clear.

Overall, the writing is strong, and these minor revisions would further enhance the clarity and flow of the manuscript.

#### **Action items.**

- **[writing]** In Section 2 (Related Work), the phrase ‘Play and Curiosity’ is used as a bolded header, but the subsequent paragraph begins with ‘Play provides...’ without a clear transition. Consider rephrasing the opening sentence to better integrate with the header or add a brief introductory clause to improve flow.
- **[writing]** In Section 3.2 (Task Proposer Team), the formula for Competence Frontier  $F(\tau)$  is presented with a parenthetical explanation of  $\bar{r}$ . The sentence structure is slightly dense. Consider breaking this into two sentences or using a colon to separate the definition from the explanation for better readability.
- **[writing]** In Section 5.2 (Cross-Environment Transfer), the phrase ‘Notable gains: cube lifting (+16.0 pp), two-arm lifting (+24.0 pp).’ is a sentence fragment. While common in technical writing, it could be integrated into a full sentence (e.g., ‘Notable gains include...’) to maintain consistent grammatical structure with the surrounding text.
- **[writing]** In the Appendix, Section A.4 (Agent Prompts), the prompt templates use a mix of natural language and code blocks. Ensure that the transition between the ‘SYSTEM PROMPT’ and ‘USER TEMPLATE’ sections is clear, perhaps by adding a brief explanatory sentence before the code block to guide the reader.